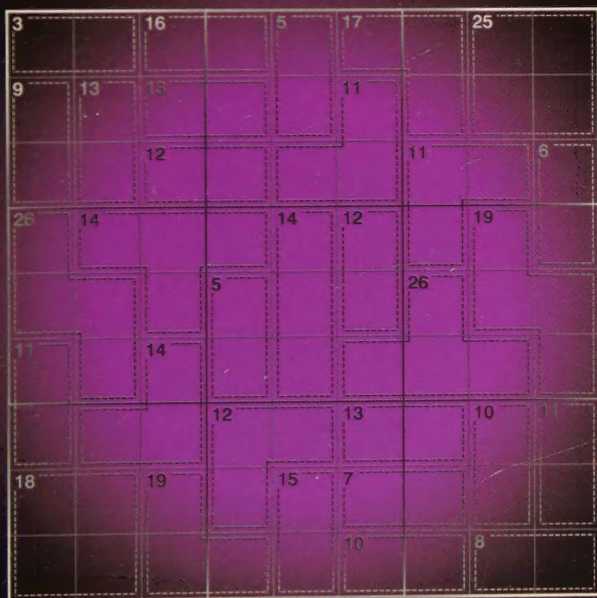


THE  TIMES

Killer Su Doku



The bestselling, deadliest Killer Su Doku series



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THE  TIMES

Killer Su Doku

The dangerously addictive Su Doku puzzle

First published in 2012 by Times Books as The Times Killer Su Doku Book 8
This edition published 2012

HarperCollinsPublishers
77–85 Fulham Palace Road
London, W6 8JB

www.harpercollins.co.uk

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ISBN 978-0-00-792841-5

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A catalogue record for this book is available from the British Library.

Layout by Susie Bell – www.f-12.co.uk

Printed and bound in Great Britain by Clays Ltd, St Ives plc.



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Introduction

How to tackle Killer Su Doku

Welcome to this edition of *The Times Killer Su Doku*. Killers are my personal favourites, and this book enables you to work your way up through the different levels, right up to Deadly.

The techniques described here are sufficient to solve all the puzzles. As you work through this book, each new level, and each new puzzle, presents some new challenges and, if you think about how to refine your techniques to overcome these challenges in the most efficient way, you will find that not only are you solving every puzzle, but also your times are steadily improving. Above all, practice makes perfect in Su Doku. In particular, with Killers, you will have to revive and practise your mental arithmetic skills.

It is also important to note that the puzzles in this book all use the rule that digits cannot be repeated within a cage. Some puzzle designers allow repeated digits, and you need to be aware of this variation in Killers, but the *Times* puzzles do not.

Cage Integration

Cage integration must always be the first stage in solving any Killer Su Doku. It is essential to get the puzzle started and, the more that is solved at this stage, the easier the remainder becomes. It will not be possible to solve the harder puzzles without doing a very thorough job of this part.

The basic concept is that the sum of all the digits in a row, column or 3x3 region is 45; therefore, if as in the top right region of Fig. 1, all but one of the cells are in cages that add up to 39, the remaining cell (G3) must be 6. Likewise, if the cages overflow the region with one cell outside, which is the case in the top right region if the cage with sum 8 is included, then as the three cages add up to 47, the cell that sticks out (G4) must be 2.

	A	B	C	D	E	F	G	H	I
1	5	13 (15)		17	17		28	11	
2		23 7 8 9							
3	15 7 8 9	7 8 9	10	3	2	9	8	6	
4	(14)	(14)	1	4	20		2	13	
5	(14)	7	20	11		18 (10)	(8)	7	
6	17					(10)	(8)	15	
7	(14)	(14)	(14)	19 (11)	20		7		16
8	8	(15)		(11)		14 (14)			
9		15 (15)		(11)		22 (14)			

Fig. 1

For the harder puzzles, it is also necessary to resolve pairs in this way as well as single cells. So, in the middle right region, the three cages plus the 2 in G4 add up to 37, leaving the remaining pair (G5/G6) adding up to 8. The individual digits in the pair cannot be resolved at this stage, so each cell is marked (8). Equally, the cage of 18 could have been included, in which case the pair that stick out (F5/F6) must add up to 10 ($2+13+7+15+18-45$). Whilst F5 and F6 cannot be resolved yet, knowing that they add up to 10 means that D4 must be 4 ($45-(10+20+11)$). This illustrates the point that solving one integration often leads to another, and it always pays after solving an integration to look and see if it now opens up another.

The bottom right region also has a pair sticking out (F8/F9) which must add up to 14 ($7+16+14+22-45$). This leads on to being able to integrate the bottom centre region, which resolves the triplet D7/D8/D9 as 11 ($45-(20+14)$). Note that the same could also be achieved by integrating column D (where $45-(17+2+4+11) = 11$).

To solve the harder puzzles it will also be necessary to integrate multiple rows, columns and regions. Two rows, columns or regions add up to 90, and three add up to 135. I remember one Killer in a World Su Doku Championship where the only way it could be started was to add up four columns (totalling 180). This is where all that mental arithmetic at school really pays off! Because it is so easy to make a mistake in the mental arithmetic, I find it essential to double-check each integration by adding up the cages in reverse order.

In this example, the cages in the bottom three rows add up to 121, revealing that the triplet in A7/B7/C7 must add up to 14.

Moving up to the top left region, the two cages add up to 18, making the remaining quartet (A3/B2/B3/C3) add up to 27. As will be explained later, there are only two possible combinations for this: $9+8+7+3$ and $9+8+6+4$. But $C3+C4$ must be 4 (because $D3+D4 = 6$), so $C3$ must be 3, the triplet A3/B2/B3 must be $7+8+9$, and $C4$ must be 1.

In the final resolutions shown in the example:

- integrating column B identifies that the triplet B1/B8/B9 adds up to 15; and
- integrating the four regions (middle left, bottom left, bottom centre and bottom right) identifies that the triplet A4/A5/B4 adds up to 14.

The integration technique has now resolved quite a lot of the puzzle, probably a lot more than you expected. To get really good at it, practise visualising the shapes made by joining the cages together, to see where they form a contiguous block with just one or two cells either sticking out or indented. Also, practise sticking with it and solving as much as possible before allowing yourself to start using the next techniques. I cannot emphasise enough how much time will be saved later by investing time in thorough integration.

Single Combinations

The main constraint in Killers is that only certain combinations of digits are possible within a cage. The easiest of these are the single combinations where only one combination of digits is possible. The next stage in solving the Killer is therefore to identify the single combinations and to use classic Su Doku techniques to make use of them.

There are not many single combinations, so it is easy to learn them. They are:

Two cell cages	Three cell cages	Four cell cages
$3 = 1+2$	$6 = 1+2+3$	$10 = 1+2+3+4$
$4 = 1+3$	$7 = 1+2+4$	$11 = 1+2+3+5$
$16 = 7+9$	$23 = 6+8+9$	$29 = 5+7+8+9$
$17 = 8+9$	$24 = 7+8+9$	$30 = 6+7+8+9$

To get to the example in Fig. 2 from the previous one:

- In the top left region, the cage of three cells adding up to 23 (B2/B3/B4) only has a single combination, which is 6+8+9. However, we already know that the triplet A3/B2/B3 must be 7+8+9. As B2 and B3 cannot contain the 7, it must be in A3, making B2/B3 8+9 and leaving B4 as 6.
- In the middle left region, the cage of three cells adding up to 7 can only be 1+2+4, and there is already a 1 in the region (C4), so B7 must be the 1, and B5/B6 must be 2+4.
- We have already worked out that the triplet B4/A4/A5 must add up to 14, but B4 has now been resolved as 6, so A4/A5 must add up to 8.
- In column A, the cage of 17 must be 8+9, and we already know that A7/B7/C7 adds up to 14 and B7 is 1, so C7 must be 4 or 5.
- The other single combination cages are: D1/D2 = 8+9, H1/I1/I2/I3 = 1+2+3+5 (note that I3 can only be 1 or 5 because row 3 already contains 2 and 3), and I7/I8 = 7+9.

	A	B	C	D	E	F	G	H	I
1	5 (15)	13 (15)		17 (8 9)	17		28	11 (1 2 3 5)	12 (3 5)
2		23 (8 9)		9 (8 9)					12 (3 5)
3	15 7 (8 9)		10 3 (8 9)	2	9		8 6		1 5
4	(8)	6	1	4	20		2	13	
5	(8)	7 2 (4)	20	11		18 (10)	(8)	7	
6	17 (8 9)	2 4				(10)	(8)	15	
7	8 (9)	1	5 4	19 (11)	20		7		16 (7 9)
8	8	(15)		(11)		14 (14)			7 9
9		15 (15)		(11)		22 (14)			

Fig. 2

Multiple Combinations (Combination Elimination)

With the harder puzzles, there will only be a few single combinations, and most cages will have multiple possible combinations of digits. To solve these cages it is necessary to eliminate the combinations that are impossible due to other constraints in order to identify the one combination that is possible.

The most popular multiple combinations are:

Two cell cages	Three cell cages	Four cell cages
$5 = 1+4$ or $2+3$	$8 = 1+2+5$ or $1+3+4$ (always contains 1)	$12 = 1+2+3+6$ or $1+2+4+5$ (always contains 1+2)
$6 = 1+5$ or $2+4$	$22 = 9+8+5$ or $9+7+6$ (always contains 9)	$27 = 9+8+7+3$ or $9+8+6+4$ (always contains 9+8)
$7 = 1+6$ or $2+5$ or $3+4$		$28 = 9+8+7+4$ or $9+8+6+5$ (always contains 9+8)
$8 = 1+7$ or $2+6$ or $3+5$		
$9 = 1+8$ or $2+7$ or $3+6$ or $4+5$		
$10 = 1+9$ or $2+8$ or $3+7$ or $4+6$		
$11 = 2+9$ or $3+8$ or $4+7$ or $5+6$		
$12 = 3+9$ or $4+8$ or $5+7$		
$13 = 4+9$ or $5+8$ or $6+7$		
$14 = 5+9$ or $6+8$		
$15 = 6+9$ or $7+8$		

Working from the last example, the following moves achieve the position in Fig. 3:

- E3/F3 is a cage of two cells adding up to 9, which has four possible combinations: 1+8, 2+7, 3+6 and 4+5. But 1+8 is not possible because 8 must be in D1 or D2, 2+7 is not possible because of the 2 in D3, and 3+6 is not possible because of the 3 in C3, so E3/F3 must be 4+5. I3 can then be resolved as 1.

- For the cage of 5 in A1/A2 there are two possibilities: 1+4 and 2+3. As the region already contains a 3 in C3, A1/A2 must be 1+4.
- The pair A4/A5 must add up to 8, which has three possible combinations: 1+7, 2+6 and 3+5. 1+7 is not possible because of the 7 in A3, and the 6 in B4 means that 2+6 is not possible, so A4/A5 must be 3+5. This also means that A8/A9 must be 2+6.
- The cage of 11 in D5/D6 has four possibilities, but all bar 5+6 can be eliminated. This also makes D7/D8/D9 into 1+3+7.

	A	B	C	D	E	F	G	H	I
1	5 1 4	13 (15)		17 8 9	17		28	11 3 2 5	3 2 5
2	4 1	23 8 9		8 9					3 2 5
3	15 7	8 9	10 3	2	9 4 5	4 5	8 6		1
4	3 5 3	6	1	4	20		2	13	
5	3 5 3	7 4 2	20	11 5 6		18 (10)	(8)	7	
6	17 8 9	2 4		5 6		(10)	(8)	15	
7	8 9 8	1	5 4	19 3 7	20		7		16 7 9
8	8 2 6	(15)		1 7 3		14 (14)			7 9
9	2 6	15 (15)		1 7 3		22 (14)			

Fig. 3

Getting good at this is rather like learning the times table at school, because you need to learn the combinations off by heart; then you can just look at a cage and the possible combinations will pop into your head, and you can eliminate the ones that are excluded by the presence of other surrounding digits.

Further Combination Elimination

	A	B	C	D	E	F	G	H	I
1	5 1	13 5	6 2	17 9	17 8		28 3	11 2	2 3
2	4 1	23 8	9 2	6 8	9 8				2 3
3	15 7	9 8	10 3	2 2	9 4	4 5	8 6		1
4	5 3	6	1	4	20 5		2	13 8	5 8
5	5 3	7 2	4 2	20 5	11 6	18 8	7 2	7 3	4 3
6	17 8	9 2	4 2		6 5		8 2	15 9	9 6
7	9 8	1 5	4 5	19 7	20 3		7		16 9
8	8 2	6 3	7 3		1 7		14 9		7 9
9	6 2	15 3	7 3		1 7		22 9		

Fig. 4

To progress from where the last example finished off to the position in Fig. 4:

- The cage of 13 in the top left region must be 2+5+6, but 2 and 6 are already in column B, so B1 is 5 and C1/C2 are 2+6. B8/B9 are then 3+7.
- The cage of 13 in H4/I4 cannot be 4+9 or 6+7, and so must be 5+8. This then leads on to the cage of 15 in the region being 6+9, which leads on to the cage of 7 being 3+4. Finally, G5/G6 are left as 1+7. This illustrates nicely the benefit of looking for how one move can lead on to the next.
- With G5/G6 as 1+7, the other side of the cage of 18 (i.e. F5/F6), which is a pair adding up to 10, can only be 2+8, because digits cannot be repeated within a cage.
- With F5/F6 as 2+8, the pair F8/F9, which add up to 14, must be 5+9.

This puzzle is now easily finished using classic Su Doku techniques.

With Deadly Killers it is also useful to identify where the possible combinations all contain the same digit or digits, so you know that digit has to be somewhere in the cage. The digit can then be used for scanning and for elimination elsewhere. It is also useful to identify any digit that is not in any of the combinations and so cannot be in the cage.

If you get stuck at any point, and find yourself having to contemplate complex logic to progress, it will almost certainly be because you did not find all the cage integration opportunities at the start. So, the best way to get going again is to look for more cage integration opportunities.

Finally, keep looking out for opportunities to use classic Su Doku moves wherever possible, because they will be relatively easy moves. Good luck, and have fun.

Once you have mastered this book, why not step up a level and try *The Times Ultimate Killer Su Doku* books?

Mike Colloby

Secretary, UK Puzzle Association

Puzzles

Moderate

11		13	3	12	18		14	9
6	14					8		
		7		15			5	22
8		17			4	10		
20			6				10	
4		24			14	19		9
8			12					
22				3		8	14	5
13		5		13				

⌚ 12 MINUTES

TIME TAKEN

20		13	15		13	19		
	10		10			8	18	
		3		16	21			
13	5					7		6
		19	6			13		
9	16		9	6		17	17	
							8	
	8		17	6	22			6
19					10			

🕒 12 MINUTES

TIME TAKEN

6	8	14	22			16	20	
			10					
16		17	13		10		18	7
4								
	14	21			13	7		15
20		3		8		16		
	19				4		12	
	9	8	6	16	7			
					12		14	

⌚ 12 MINUTES

TIME TAKEN

Moderate

6	17			16		16		12
	20				22			
14	8			21	8			
		13	26			25		
27					8		8	
				11		8		23
21							17	
	23		8		18			
						9		

⌚ 12 MINUTES

TIME TAKEN

20		17			21			21
10			17	14				
	16	12				4		
6				8			24	
	14		15	5	10			
16	17	3			22		9	13
			8		8			
		12		16			8	4
8		10			17			

🕒 15 MINUTES

TIME TAKEN

Moderate

8	16			26	20			6
	22				23			
		14	19					13
20	10					4		
			19			12		
		16	12	4		17		23
8					10		7	
9			21	14				
15				8			9	

⌚ 15 MINUTES

TIME TAKEN

15		12		6			28	17
4		14						
12	13	7		21				
		16	11		11		4	
11			4		17		15	7
	10		21					
10	24		13	15		11	3	
		5		5			11	17
			7		8			

⌚ 15 MINUTES

TIME TAKEN

19	23	19			11	8		
			8			19		
		21		16	3	16		17
	18					11		
			25					
3		8		8	20		11	13
10	6				12			
		18					17	
11		8		16		10		

⌚ 15 MINUTES

TIME TAKEN

19		12		8			14	
		9	20			6	24	
15	6			7	12			
		23	13			16		
11						16	8	
19			13	20			13	
					12			15
9		8	14			13		
12				9			9	

 20 MINUTES

TIME TAKEN

28	21				16		12	
				12				27
19		19	3		5			
8				13	17			4
	3				16	11	15	
15		7	13					16
	17			7		12	3	
5		10		19				
	15				4		13	

 20 MINUTES

TIME TAKEN

Tricky

19		14		9	14		12	
		24			9		23	
7			12	6	12			
17	28					23	7	
			11					12
			20			19		
				22			18	
21	13							
			10		23			

⌚ 25 MINUTES

TIME TAKEN

23				7		10	14	
21	21			9			21	
	15				23		13	
			18					
16				14	12		20	
	4		14		6	13		
	22						12	
6		16	20			21		
				8			6	

⌚ 25 MINUTES

TIME TAKEN

8		10		29	10	9	20	
28	16						16	
				11				
	3				17			22
16		22	21			22		
			3	5	10		13	
12								
20		4	13	17		7		
				8			13	

 25 MINUTES

TIME TAKEN

21		18		18		24		
			23					16
		11		18	19			
22							23	
		8		15	15	13		
12		12					9	
20						22		16
		18						
13		19						

⌚ 25 MINUTES

TIME TAKEN

3		22		9		10		15
14		24		18				
13					7		19	
			7		22	13		
23		13					19	
			12		23			22
17	8			23				
	17	3						
			9		20			

⌚ 32 MINUTES

TIME TAKEN

34		21			12		3	
			24				30	
7			18					9
10	20			21				
						21	10	22
23	10	11	20					
			13					
	6	13			3		17	
		16		11				

⌚ 32 MINUTES

TIME TAKEN

3	19	14		7		9	15	
		11	20		8		20	
16				18				
		12			28	18	7	
			10					18
23		18				21		
14		28		4		6	14	
9				9			6	

⌚ 38 MINUTES

TIME TAKEN

17		10		23			6	
		19	11	3		21		
20				15		5		13
		9	19			22	9	
								11
23			14		12	5		
	16	17					12	
16			19		16			22

⌚ 38 MINUTES

TIME TAKEN

26	24	16			20			
				8		7	10	
	23	11			14			18
			23					
4		8				21	10	
	7	13		12			22	16
18			20		20			
		22						
						12		

 38 MINUTES

TIME TAKEN

25		14			26		15	
12			4			16		
	11		20	11			23	
21	11							
					16	23		8
		12	17	17		3		
22							6	
				13		15		23
6		15						

 40 MINUTES

TIME TAKEN

Tough

13		11			11		10	14
6		20	23					
			15	18		18		9
11		7			6			
20			8		11	9	17	
	15	16		9			19	14
7	8		13		23		13	
	11							

 42 MINUTES

TIME TAKEN

15	23	4	25		20		19	
					7			
11		24		11		7		31
9			22		16	23		
3	11			16			17	8
28			13		17		8	
10						7		

⌚ 42 MINUTES

TIME TAKEN

5		22	22		21			7
12					10			
22			6	13		19		
10		17			21		8	
				21				18
21	18					3	6	
	7			13				
	16	4		12	14	7	13	17

⌚ 42 MINUTES

TIME TAKEN

20		8		11		21	20	
		11		17				
9	19		23				13	
							22	
18	16		7	24				
		10				32		
				11				10
28	12			17				
			19			7		

⌚ 42 MINUTES

TIME TAKEN

8	19		16		12	6	19	
			9					4
17		12		8		21		
			10	21			13	
16		15				10		
10			7		15		21	
17			8				17	
4		14		21				
18					17			

 42 MINUTES

TIME TAKEN

21				20				8
30					20		23	
18			11			19		
	13	10			21		11	
8								22
		20	16	10				
					13	17	7	
17							13	
	22			15				

⌚ 42 MINUTES

TIME TAKEN

27	21				20			
	23	20	9		3		27	21
			19					
4		12		13			21	
	9	19	12		25			
6			17				13	
					23			12
13		12				4		

⌚ 44 MINUTES

TIME TAKEN

Tough

33				14	20	20		
12							20	
	13		21					
	32				4	18		
18			19			20		
					32			13
	20					28		
19			12					8
					9			

⌚ 44 MINUTES

TIME TAKEN

7	22		17				17	
			10	21				
23	8			19				22
			22	20		21		
17				8				
	20		11			8		
	18				9		14	
15				22		8		9
					17			

⌚ 44 MINUTES

TIME TAKEN

23		24					17	
	13	21			7			
		17	23	12		9		22
13						15		
	22		21		8			
						21	16	
13		20						
	22			7	20			19

⌚ 44 MINUTES

TIME TAKEN

20	10		20		12		17	
	14			13		16		
		9	16	8		9		21
	14							
12		19			18			
			7				19	
12		10			17		8	
8	21		9	15	11			
						20		

⌚ 44 MINUTES

TIME TAKEN

16		8			21		16	
		30						18
21		12			22			
			20			8		
21					12			
17				6		9	16	
13	8		28				20	
				7		15		
20				12		9		

⌚ 44 MINUTES

TIME TAKEN

30	15		26		11		7	
						12	23	
		8	20					
32				8		19		
		7	26			25		
					17			29
15	20							
		18			16	14		
			7					

⌚ 44 MINUTES

TIME TAKEN

Tough

29		21	21				21	
					19			12
10		9		7				
	22		21			9		
6	9				17			17
		14				15	8	
8		18		16	12			
22							22	
				20				

⌚ 44 MINUTES

TIME TAKEN

13	17		7			16		
		9		19			22	
19		5	11	20			20	
						15		7
17		11	17					
	20		29		19		18	
								16
13			23		8	5		
						9		

⌚ 44 MINUTES

TIME TAKEN

16	18			9	20		11	
		8	20			8	25	
9							17	
10	17				11			
	30	23		10			12	
			14		12	7		15
							14	
16		10				21		
			15				7	

 44 MINUTES

TIME TAKEN

10		3		21			11	
23				23			10	
9		9		16				
17	7	20		9		10	30	
		30		19				
							20	17
9				8				
14		14	15		10		21	

⌚ 44 MINUTES

TIME TAKEN

Tough

23				17		21		14
7		8						
19		8	24	17		9	20	
				17				
11	12						21	
	11	20			14	17		6
			8					
	21	7		16			14	
		12			8		3	

⌚ 44 MINUTES

TIME TAKEN

26		19			13	15		8
	19							
		19	7		28	15		14
18						7		
	11	7			21		26	
		12		6				
12		3					11	11
22			19		7			
	10				19			

⌚ 52 MINUTES

TIME TAKEN

Tough

17			12		31			
	8		21	17			19	
41	20			9	20			
						18		
								14
		22		17				
11				20			29	
9		13				15		
	17		5					

⌚ 52 MINUTES

TIME TAKEN

23	19		11		28		15	
	9							15
		8	22	20	15			
							23	
	24				12			
		11			26	7	8	
11							22	
23		14			12			
		19					8	

⌚ 52 MINUTES

TIME TAKEN

4		23			16	16		
16			11				22	
24		17				24		
				7				21
15		18						
28		12		19	13		12	7
		17						
						30		
11		11			11			

⌚ 52 MINUTES

TIME TAKEN

11		4		28			20	
23					20			
12			30	25		17		
	13							7
20			19					
			7			21	20	
	22	12		6				
					11		28	
	11		14		4			

⌚ 52 MINUTES

TIME TAKEN

20			22		12			
15					21			
4	7	10		6		24		
		12		13		21	23	
12	16		18	6				
	9	9					7	
				27	7		19	
15						22		9
8		11						

⌚ 52 MINUTES

TIME TAKEN

21			18				9	
	16			26			22	
10		21	12			20		
					8			
24						13	16	15
20	23		5		11			
	7		19				7	
				22				26
14								

⌚ 52 MINUTES

TIME TAKEN

Tough

22	24		18	6		19		
				9	18		3	
	18					20		
			10		20		25	10
6	14							
	23	10			15		10	19
		12				16		
8	19			15				
				7		9		

⌚ 52 MINUTES

TIME TAKEN

14		16	23	10	14		10	27
					16			
7				21				
9		8				8		
26		9		19				15
	17		11	12				
				10		12		3
16		13		21				
	20				9		9	

⌚ 52 MINUTES

TIME TAKEN

Tough

21	24			7		20	11	
	8		16		14			14
20		19	20		9		9	
						12	19	
18	6		18		21			
		16				20		10
8		12						
12			21					

⌚ 52 MINUTES

TIME TAKEN

14		11	8		17			21
29				15				
21			7			27	17	
			16					
		22						
20				8	14		7	27
	29				11			
10			23		7	18		
		6						

⌚ 52 MINUTES

TIME TAKEN

15				23	10	28		7
7			11					
12	20					6		30
		22			18			
19			23					
	18	19				20		21
							8	
			9		21			
15		16					7	

🕒 52 MINUTES

TIME TAKEN

Deadly

18		17		11		11		22
			20		16	8		
25							24	
			17					
	23			26		9		
19						19		4
		10		30	13		12	
10	15					26		

⌚ 55 MINUTES

TIME TAKEN

13	6	16		22	16	23		
							7	
11	24	7				14	19	18
				23				
14		8						
	22			10		18	14	
	9				8		9	
22	17						15	
		3		17				

⌚ 55 MINUTES

TIME TAKEN

29				11		7		20
	19	20				19		
9		15						
			20			27		
19		17		17			5	
12			23				16	
				19		12		
4		12				25		
13			11			4		

⌚ 55 MINUTES

TIME TAKEN

18	8			20			7	12
		20		28				
6					4	26		14
23	30	10					4	
		20	20		17			
			16			8	19	
								13
	12			21				
		21				8		

 1 HOUR

TIME TAKEN

26		7		10	23		21	
12								
	10		30		12		5	
10				8	18		15	
		21			10		12	
7	30		8					19
				22	10		10	
					12			
20						17		

⌚ 1 HOUR

TIME TAKEN

25	14	10	15	10			24	
								15
		21		25	13		22	
			6					
20				10			18	
10		8	16		18	13		10
30			12		10		14	
				7			9	

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

22		16			17	22		
		15					3	
14	23			21		20	16	
		13					7	
	10							16
21		12	3	9	20		19	
	14	22			12	6		6
		10				16		

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

17	20		23	19		23		
				7				
20	27				15	13		25
			18					
					25			16
12	12		21			8		
		20	11		13		23	6
11								

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

20				8		29		7
19		23						
		29			27		10	
16						23		
			20		8			35
29		17				27	14	
	16			12				
		16						

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

20	18	19			12		20	
		9			9			
			28			22		8
	29		7					
17		10	20				22	
			9	15		8		
	15				16		10	
		17		8			17	9
					11			

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

33					6		21	
22		7						21
19		8		28		23		
	9		16					
	9					22	21	
		20	12	13				
9								
11				17		18		
	25				15			

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

16	8	14			9		19	21
			11		18			
	16			21				
7	20					23		23
					20			
15	25			18		6		
		10				13		
			22				18	
22					10			

 1 HOUR 5 MINUTES

TIME TAKEN

25		20		11			9	
				9			19	
18		30			17			11
		7					9	
12	20		11		21	9		
							17	
25		11				21		11
	18		11	22				
							11	

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

8		11		19		9		21
22					26			
8		17		8			7	
19			18	11				
				23		21		
	19		22		17		10	
12								
	8			17		10	17	11
		14						

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

11	4	18			20			
		22			8		19	
	30					19		
12		7	11	17	11		12	
								17
19		17		11		15	13	
					23			
17	11	12						16
				13				

 1 HOUR 5 MINUTES

TIME TAKEN

9		17	12		22			
				20	16		22	
22	13				23		9	
		10						7
8	20							
		9	13		14	24		22
7	27		16			8		
				11				
		10		3		11		

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

9	17		8		18			
	15	24			28			9
				15	16	13		
6							8	
22		17						18
	20		9		6	14	4	
19			24					
		15	7		10		23	
						11		

 1 HOUR 5 MINUTES

TIME TAKEN

30				7		7	23	
12	13	19						13
			19	8		17		
18				30			16	
	21							
			19			11		18
20		7			10			
11		20			10			
				18			8	

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

12		7			25		27	
23		10	20				9	
	18				8	19		
8				35				12
		13						
	16		28		12		22	
13					6			
		16		6	13		13	4
					10			

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

19		8			12		11	20
		18	20					
12	15		16		17			
						22	11	
		19	28		20			3
14	7					7	5	
					24			11
23			10			23		
							10	

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

30	11				22			16
		19				10		
8		9		15			16	
			22		13	10		
16	13	17						
			7			23		
8		15		17			9	
	22	9				20	18	
			10					

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

13	7			17		20		
	21	9	21				10	
				14		22		14
	29						10	
10				23				
7			13			20	13	22
16	13		8		20	8		11
		14						

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

15		13			17			
7		28			14	19		
20			9				16	22
	21			10				
		21			12			
	19	11				12	13	
			13				3	
8		18		8	23		10	10
13								

 1 HOUR 5 MINUTES

TIME TAKEN

17		9		20				17
11		23		18				
			9		23	19		
18			15					
				10	7	23		
13	22		18				6	
	8			16		13	22	
					8			
15		15				10		

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

4	22		27		12		19	
			9			20		
22		10			17			
				29			23	
10								23
21	10	11		20	18		17	
			15		23			
7						16		

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

20		11			22	15		
11		12	19					8
				16		19	9	
4	23		20					
					3		23	
22			12	19		12		
	18							
14				15	20			12
		7			19			

⌚ 1 HOUR 5 MINUTES

TIME TAKEN

6		29			8		24	
20			18	7				
		22		13	26	13		24
9								
		10				12		
27			8			17	19	17
			8					
8		11		23	13			
13								

 1 HOUR 10 MINUTES

TIME TAKEN

3	29		13	8		17		21
9		10	24	20				12
22							25	
	17		14	21		9		
								4
27				21	30		12	
	9	9						10
					9			

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

17				18			13	
15		30						
	24		17	27	21			
						25		
12						7		
28	24		12	14		23		10
				4				
					15		20	
29								

 1 HOUR 10 MINUTES

TIME TAKEN

11	8		19		17			16
	16					14		
20		16	8		22	17	19	
		24	13					
			23		6		20	
20		11		17		4		
			12				19	
14				7		12		

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

4	25					24		15
	25			19				
18		13				13	18	
	11		13	17				
				8	13			16
21		14				15		
	14	26			13			
			20			11	19	

 1 HOUR 10 MINUTES

TIME TAKEN

21		21				34		
		10	12		7			
18			21			8		
		20		20		21		
7							9	
20		11		19	17			21
12		9				22		
		14	19					
12								

 1 HOUR 10 MINUTES

TIME TAKEN

24		14		22	10		17	
		23						
			11	5		23		9
	20				28			
			21				22	
4	8			22		10		
			20				16	
22	15							20
		6			13			



1 HOUR 10 MINUTES

TIME TAKEN

9	10			13		25		
		21		20	11		17	
17						8		
	25			11		17		
20					8			10
	20		16	8		13		
8					23			
	15	21		3		18	18	

 1 HOUR 10 MINUTES

TIME TAKEN

17	21		6	14			8	
				22			20	13
	21		19		17			
9		19						
					34			
7		19		12			19	
15				19				17
	17	12				9		
			19					

 1 HOUR 10 MINUTES

TIME TAKEN

12			25	10		19		12
	29							
		20	22	11				9
15				17				
					12		26	
14		19		29	15			
16						4	15	
	7	16			14			
							17	

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

18	16		16		13		8	
				24		19		
15	14		28				17	
				19		20		
18						8		
19		12		8			11	
					23		18	
	10	11	3			11	15	
			11					

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

26	3		19		24			12
		21				13		
	8			19		19		
		20					16	22
16	18			22				
			16		20			
	13					22		
21		16						12
					7			

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

22	8	20	23	17			23	7
					17			
	19		21			7		11
3				20				
							23	
11		9		20	14	12		17
6			13					
11	25					7		11
					8			

 1 HOUR 10 MINUTES

TIME TAKEN

20	19			16				22
			26		10			
21					10	21		
17			14				21	10
13		20						
			8	17	17			
22							11	
14		6			13		24	
			13					

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

10		3		22		10	19	12
17		22						
			24		13			
8		20			8	23		
	19						17	10
22		18		10				
					12		17	
	18				11	3		16
6		15						

 1 HOUR 10 MINUTES

TIME TAKEN

10	20	24			18			13
		20				12		
12			7		21		11	
		8			10			10
6	20		22					
	17			18	20		14	
							19	
17		11	11		8			
				5		21		

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

18	8		23		16	18		
		20				10		17
			16		28		10	
8				15		12		
	24							
19		19			19	17		
				4			12	
	21					28		
		18			5			

 1 HOUR 10 MINUTES

TIME TAKEN

15		18		8			18	
17				28				
	11	23	11	7			23	21
					13			
10				22		10		
17	18		17		4		19	9
						12		
	18		9					
			18				9	

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

20				4	22			7
26					14			
13			13	21	18		17	
	18					3	23	
		19						
24				20		23	22	11
18		19						
		10			8			
							12	

 1 HOUR 10 MINUTES

TIME TAKEN

21		17		17		9		8
		8			12			
22			22	3	10		16	
					19		7	
11			12				11	22
17	20	10		9	12			
		16			5			
		10	24				27	
			8					

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

18			19		13	19		
20	7							22
		15		15	10		10	
	25							
		11			25		17	
8			18			21	7	
	13							10
18			13		19			
10				22				

 1 HOUR 10 MINUTES

TIME TAKEN

9	21		15			23		
			10				17	
24			21		21			
	23		21				17	
12								11
	7	20	17		23	17		
			11			20		
13		22			10			

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

20		14				15		11
		22			26			
21				11				
21						13		24
31					20			
16	12		18			31		
					15		19	
10	20			15				

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

7	17		12		20			
				18	22		12	
20		20				11	23	
25			22					7
					12			
	21		18				9	30
				29				
10					4			
	8		12		16			

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

14		19		7	11	20		11
8								
19	21	19		16		12	5	14
		13	22		15			
					14	26		17
20	8		13					
					21	7	13	3
	17							

 1 HOUR 10 MINUTES

TIME TAKEN

20	18			21			3	
		22		7			26	
7					23			
	21		21					13
20					18			
			9		9	21		13
20	8		28					
		22				15		
					16		4	

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

14	26			20		5		28
		18	10			29		
					29			
26			13	26				
						19		
22		19					9	
						17	23	
23								
		20				9		

 1 HOUR 10 MINUTES

TIME TAKEN

22		27	13				18	
			17	12	15			
					12		27	
28			7		21			
		3		24				
23	17				21	10		
				16				
	7		23		20			
					22			

🕒 1 HOUR 10 MINUTES

TIME TAKEN

16			30				8	
21		22		27				
			9		20		22	
	21		20		31			
16			18				32	
	11					3		
							16	
	23	29		10				

 1 HOUR 10 MINUTES

TIME TAKEN

28		18		18			18	
				29				
8	19			8				23
	8					19		
	30		21		21			
26						22		
				25			16	
	28					20		

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

15		10		30		10		22
8		27						
	29			14		13		
7				16		16	20	
		6						
20			26			13	17	
		25		19		24		7
11								

 1 HOUR 10 MINUTES

TIME TAKEN

26			8	27		22		22
	20				7			
17							23	
		8						
22			11		26		18	
	9	35	20					11
						13		
				10		8	20	
11		11						

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

26	8			26	19			6
	19				15			
			6	31	16			12
	21					14		
							28	
	11		14		22			
8			15			12		
29					12	24		
	11							

 1 HOUR 10 MINUTES

TIME TAKEN

10	20		22		16		14	
				17			21	8
25		10						
			8	26				
15					21		19	15
	10			15				
3	30	8					8	
		12			12	12		18
		10						

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

24			6		21			
23				37	15		22	
12						13		
	21				18			22
		11				22		
20	16		20					
							22	11
		10		15				
9				15				

⌚ 1 HOUR 10 MINUTES

TIME TAKEN

16		9		21	16		6	
					22			
13	17	23		28	18			7
			13		8			
					12		8	
12	12		7			14		17
				16				
11	16	12			16		19	8
		8						

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

7	22			25		11		
		11	10				20	
29			6		28			
				29			23	29
9	16				11			
		11		21	10			
						9		
20		23					8	
				17				

 1 HOUR 12 MINUTES

TIME TAKEN

24	8		17		15		25	
		19					6	
		13		17		22		
11		11						
21			15		13		9	
		18		8	20	12		8
10						22		
	8	11	16					
				11		15		

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

15		23					13	
9		16			21			
18			14	22		23		5
	13							
			16		8		26	
	29		29			16		22
18					17			
	12		9					
							11	

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

19			8		22			7
26			24		8			
			28		29	13	16	
11	32		19				29	
				11		10		
		24			15			
						24		8
		22						

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

20	3		30	12			21	
				22	18			
			20		21		27	
11								7
29		7						
		24	20	24		16		
							26	
12						8		
		7		20				

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

23				7		18	8	
40		11			22			16
				3			35	
27		16						26
				31				
11		18						
			34				12	
14		13			20			

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

36			35		15			
		22						17
			23		17			
21			7				20	
		11		18		31		
			23			33		
9	28							8
			15			12		
							4	

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

9	17	27					9	
		20				14		
		23		23			23	
21				10	14			
		3			8		17	
7	19		11	19	18			
	11						13	
19		18			9	7		16

⌚ 1 HOUR 12 MINUTES

TIME TAKEN

19			9	13			14	
	28				9		20	
7				13			16	
22			17		9			
8				13		20		11
24		13			25		3	
								14
11			15	30			12	
6				4				

 1 HOUR 12 MINUTES

TIME TAKEN

45				25		23	19	
			17				8	
25				11				
					7		8	
21		12				32		
			28				31	
	39	19						
						22		
			13					

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

25		16		12		23		6
			17			15		
21			22					18
	20	11		16				
			8		21			
7					22	19		
	23						8	
	18		23					
		19				15		

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

14	30			30	8		10	
	25							
		5	27	19		33		
							21	
				12				
28						19	8	
18		21					16	17
	18				16			
			10					

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

9	28	16		15			7	23
				22				
	16		7		12	18		
21				19			5	19
14		11		15	24	24		
			21					
23							9	
10						17		

 1 HOUR 15 MINUTES

TIME TAKEN

16	11		8			16		24
	17		16	23				
	23						13	
				20				24
7		18			23			
26								22
				17				
13	12		27	3		12		
						14		

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

10		21		23			9	
		23			16		20	
30				19				10
		19				11		
			12		21		7	
7						19		
11	23	20					13	
			8	19	20			
							14	

 1 HOUR 15 MINUTES

TIME TAKEN

8		19			12		22	
	22				27			
24		22	19	24				
27					9	13		
							7	30
				35	14			
28					20			
		4			19			

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

21	16	19	26			12		
			7				27	
	21			10		15		4
			15					
7				23		16		16
8			11		12	13	12	
30				4				
			24				20	12
4								

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

25	18	21		19	11		20	
					21			
			12			12	18	
	22	7		22				
9						8		
				9	20		19	
21					4			
31			28			6	9	
							13	

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

24	17		5		25	7	21	
		40						
24					9		19	
17			17		25			
7			17				32	15
	18							
7		35			8			
						16		

⌚ 1 HOUR 15 MINUTES

TIME TAKEN

15		8	19		14			20
23						23		
	16		15				14	
				20	13	11		
20							25	
				16		12		
6	13		30	12			20	
	26							
				4		10		

🕒 1 HOUR 15 MINUTES

TIME TAKEN

20		29					15	
		18	4		9		9	
13	20		22	20				
					5		13	
8		11		18			7	
	26			20			34	
					3	22		
12		15						
		18			14			

⌚ 1 HOUR 25 MINUTES

TIME TAKEN

22		19		24			36	
					21			
8			13					
16					19	19	9	37
20			23					
14		20						
				34		25		
	19							
		7						

⌚ 1 HOUR 25 MINUTES

TIME TAKEN

12		24				22		
16			16	26		12		
	18			17			8	
					21			14
22		20				13		
					32		12	
24								12
		21				3		
17					23			

⌚ 1 HOUR 25 MINUTES

TIME TAKEN

13	15		16		24	21		6
		12					19	
13			23					32
					26		7	
13				16				
17	21				23	21		
				13			22	
		10						
22								

⌚ 1 HOUR 25 MINUTES

TIME TAKEN

23		21			42	9	3	
		29					35	
			3		12		11	
20						17		
		8						
19		22		28		8	23	
			8					7
8	11							
		20				18		

⌚ 1 HOUR 25 MINUTES

TIME TAKEN

19	32				17			18
		21		31				
			19		7	9		
15							7	
20		29				12	21	
					27			
	18	19	17				28	
								8
					11			

⌚ 1 HOUR 25 MINUTES

TIME TAKEN

31				12			20	
		30						
19			12			18		19
	32			11				
			19		10			
10					29		29	
20		10	15					
				23		10		
13							13	

⌚ 1 HOUR 25 MINUTES

TIME TAKEN

17		8		20	23			24
20		24						
			17		15			
							20	
29	9	10			23			
			20			11	8	
	18						30	
				24				
15			20					

⌚ 1 HOUR 45 MINUTES

TIME TAKEN

15				15		15	32	
18		20	18					
					17	19	4	
26								
		22	31				21	
13						33		
		24		24				15
	15							
				8				

 1 HOUR 45 MINUTES

TIME TAKEN

12	26		35				7	
						12		18
	28		32		10		21	
30				34				
	18						25	
			22					11
12		12				10		
		14				16		

 1 HOUR 45 MINUTES

TIME TAKEN

14			32					
	23	32			17		37	18
				31				
5		26	27					
					6			
22				11		32		
	17							13
			19	23				

 1 HOUR 45 MINUTES

TIME TAKEN

28		22		19	12			23
12								
	17		19	26				29
				19				
10						12		
	30		9					
9				14	19	22		
22						5		15
	12							

⌚ 1 HOUR 45 MINUTES

TIME TAKEN

22	21					21	
		18	32	12			19
20			19			22	
				17	11		
20	13	12			15	21	
		21			17	10	
							9
	7		13		13		

⌚ 1 HOUR 45 MINUTES

TIME TAKEN

36	7		24		23	17		
25			20		19	8		
							24	11
	18			21	24			
	9							
		21	12	19		11		
15					12			22
		7						

⌚ 1 HOUR 45 MINUTES

TIME TAKEN

16				24			24	
7	21	16						
		24	18		28		26	
						23	8	
28								
		20			15		3	
	24						19	
9						15	18	
	19							

⌚ 1 HOUR 45 MINUTES

TIME TAKEN

23				22	17		9	
10		21						14
21			16			25		
				7				
		11		21	11	32	13	
19								
		13		19				8
11		20	21			21		

⌚ 2 HOURS

TIME TAKEN

21	10		18	18	28	16		
							20	
	27					7		
20					21			
			9	21			23	
4		30			23	10		
7								
	16		23		16			10
				7				

⌚ 2 HOURS

TIME TAKEN

28		8			19		26	
18			39					
	20				11			
					19			
	11		19		23	7		16
10			12			21		
35								
		9		20			14	
			20					

⌚ 2 HOURS

TIME TAKEN

**Cage Reference
and
Solutions**

Cage Combination Reference

These are the various combinations of numbers for every possible total covering each of the cage sizes. This should save you time when checking the possibilities.

Cages with 2 squares:

Total	Combinations
3	12
4	13
5	14 23
6	15 24
7	16 25 34
8	17 26 35
9	18 27 36 45
10	19 28 37 46
11	29 38 47 56
12	39 48 57
13	49 58 67
14	59 68
15	69 78
16	79
17	89

Cages with 3 squares:

Total	Combinations
6	123
7	124
8	125 134
9	126 135 234
10	127 136 145 235
11	128 137 146 236 245
12	129 138 147 156 237 246 345
13	139 148 157 238 247 256 346
14	149 158 167 239 248 257 347 356
15	159 168 249 258 267 348 357 456
16	169 178 259 268 349 358 367 457
17	179 269 278 359 368 458 467
18	189 279 369 378 459 468 567
19	289 379 469 478 568
20	389 479 569 578
21	489 579 678
22	589 679
23	689
24	789

Cages with 4 squares:

Total	Combinations
10	1234
11	1235
12	1236 1245
13	1237 1246 1345
14	1238 1247 1256 1346 2345
15	1239 1248 1257 1347 1356 2346
16	1249 1258 1267 1348 1357 1456 2347 2356
17	1259 1268 1349 1358 1367 1457 2348 2357 2456
18	1269 1278 1359 1368 1458 1467 2349 2358 2367 2457 3456
19	1279 1369 1378 1459 1468 1567 2359 2368 2458 2467 3457
20	1289 1379 1469 1478 1568 2369 2378 2459 2468 2567 3458 3467
21	1389 1479 1569 1578 2379 2469 2478 2568 3459 3468 3567
22	1489 1579 1678 2389 2479 2569 2578 3469 3478 3568 4567
23	1589 1679 2489 2579 2678 3479 3569 3578 4568
24	1689 2589 2679 3489 3579 3678 4569 4578
25	1789 2689 3589 3679 4579 4678
26	2789 3689 4589 4679 5678
27	3789 4689 5679
28	4789 5689
29	5789
30	6789

Cages with 5 squares:

Total	Combinations
15	12345
16	12346
17	12347 12356
18	12348 12357 12456
19	12349 12358 12367 12457 13456
20	12359 12368 12458 12467 13457 23456
21	12369 12378 12459 12468 12567 13458 13467 23457
22	12379 12469 12478 12568 13459 13468 13567 23458 23467
23	12389 12479 12569 12578 13469 13478 13568 14567 23459 23468 23567
24	12489 12579 12678 13479 13569 13578 14568 23469 23478 23568 24567
25	12589 12679 13489 13579 13678 14569 14578 23479 23569 23578 24568 34567
26	12689 13589 13679 14579 14678 23489 23579 23678 24569 24578 34568
27	12789 13689 14589 14679 15678 23589 23679 24579 24678 34569 34578
28	13789 14689 15679 23689 24589 24679 25678 34579 34678
29	14789 15689 23789 24689 25679 34589 34679 35678
30	15789 24789 25689 34689 35679 45678
31	16789 25789 34789 35689 45679
32	26789 35789 45689
33	36789 45789
34	46789
35	56789

Cages with 6 squares:

Total	Combinations
21	123456
22	123457
23	123458 123467
24	123459 123468 123567
25	123469 123478 123568 124567
26	123479 123569 123578 124568 134567
27	123489 123579 123678 124569 124578 134568 234567
28	123589 123679 124579 124678 134569 134578 234568
29	123689 124589 124679 125678 134579 134678 234569 234578
30	123789 124689 125679 134589 134679 135678 234579 234678
31	124789 125689 134689 135679 145678 234589 234679 235678
32	125789 134789 135689 145679 234689 235679 245678
33	126789 135789 145689 234789 235689 245679 345678
34	136789 145789 235789 245689 345679
35	146789 236789 245789 345689
36	156789 246789 345789
37	256789 346789
38	356789
39	456789

Cages with 7 squares:

Total	Combinations
28	1234567
29	1234568
30	1234569 1234578
31	1234579 1234678
32	1234589 1234679 1235678
33	1234689 1235679 1245678
34	1234789 1235689 1245679 1345678
35	1235789 1245689 1345679 2345678
36	1236789 1245789 1345689 2345679
37	1246789 1345789 2345689
38	1256789 1346789 2345789
39	1356789 2346789
40	1456789 2356789
41	2456789
42	3456789

Cages with 8 squares:

Total	Combinations
36	12345678 41 12356789
37	12345679 42 12456789
38	12345689 43 13456789
39	12345789 44 23456789
40	12346789

1

8	3	7	1	4	9	2	5	6
4	5	6	2	8	7	1	9	3
2	9	1	6	3	5	7	4	8
6	2	9	8	7	3	4	1	5
5	7	8	4	2	1	6	3	9
3	1	4	9	5	6	8	7	2
1	4	3	5	6	8	9	2	7
9	8	5	7	1	2	3	6	4
7	6	2	3	9	4	5	8	1

2

3	1	5	7	8	2	9	4	6
9	4	8	1	6	5	3	2	7
7	6	2	3	9	4	5	1	8
8	2	1	6	7	9	4	3	5
5	3	9	2	4	8	6	7	1
6	7	4	5	3	1	2	8	9
1	9	6	4	2	7	8	5	3
2	5	3	8	1	6	7	9	4
4	8	7	9	5	3	1	6	2

3

4	3	8	7	6	9	5	1	2
2	5	6	3	1	4	7	8	9
9	7	1	8	2	5	3	4	6
3	9	7	5	4	2	8	6	1
1	6	4	9	8	7	2	5	3
5	8	2	1	3	6	9	7	4
7	4	9	6	5	3	1	2	8
8	2	5	4	9	1	6	3	7
6	1	3	2	7	8	4	9	5

4

4	6	3	8	7	9	2	1	5
2	9	7	3	1	5	8	6	4
8	1	5	2	4	6	9	7	3
1	5	4	6	8	2	7	3	9
7	8	2	4	9	3	6	5	1
9	3	6	7	5	1	4	2	8
5	7	1	9	6	4	3	8	2
3	4	8	5	2	7	1	9	6
6	2	9	1	3	8	5	4	7

5

2	1	3	9	5	7	6	8	4
6	8	9	2	3	4	7	1	5
4	7	5	1	8	6	3	9	2
5	9	4	3	6	2	1	7	8
1	6	8	7	4	5	2	3	9
7	3	2	8	1	9	5	4	6
9	4	1	6	2	3	8	5	7
8	2	7	5	9	1	4	6	3
3	5	6	4	7	8	9	2	1

6

4	5	8	3	7	2	6	9	1
1	7	2	4	6	9	8	3	5
3	9	6	1	5	8	2	4	7
8	6	5	2	9	7	3	1	4
9	4	3	6	8	1	5	7	2
2	1	7	5	3	4	9	8	6
5	3	9	7	1	6	4	2	8
6	2	1	8	4	3	7	5	9
7	8	4	9	2	5	1	6	3

7

9	6	5	7	3	1	2	8	4
1	3	8	2	4	6	9	5	7
4	7	2	5	9	8	1	3	6
8	2	9	6	5	4	7	1	3
5	4	7	3	1	9	8	6	2
6	1	3	8	7	2	4	9	5
3	9	6	4	8	7	5	2	1
7	5	1	9	2	3	6	4	8
2	8	4	1	6	5	3	7	9

8

5	8	2	7	9	6	1	4	3
7	6	1	4	3	5	9	8	2
3	9	4	1	8	2	6	7	5
4	5	8	6	2	1	7	3	9
6	7	9	8	5	3	4	2	1
2	1	3	9	4	7	8	5	6
9	4	5	3	1	8	2	6	7
1	2	7	5	6	4	3	9	8
8	3	6	2	7	9	5	1	4

9

9	7	4	8	5	2	1	6	3
1	2	6	3	9	7	4	8	5
8	5	3	1	4	6	2	9	7
7	1	9	4	3	5	6	2	8
5	6	8	2	7	1	9	3	4
3	4	2	6	8	9	7	5	1
2	9	5	7	1	8	3	4	6
6	3	7	5	2	4	8	1	9
4	8	1	9	6	3	5	7	2

10

2	3	1	9	8	6	7	5	4
7	6	9	4	5	2	1	3	8
8	4	5	1	7	3	2	9	6
5	7	6	2	1	8	9	4	3
3	2	8	5	4	9	6	7	1
9	1	4	6	3	7	5	8	2
6	8	3	7	2	5	4	1	9
1	9	7	3	6	4	8	2	5
4	5	2	8	9	1	3	6	7

11

1	2	6	8	3	9	5	7	4
7	9	5	2	6	4	3	8	1
3	4	8	7	5	1	2	6	9
6	7	9	5	1	2	8	4	3
2	8	1	4	7	3	6	9	5
5	3	4	9	8	6	1	2	7
4	5	2	1	9	8	7	3	6
9	6	7	3	2	5	4	1	8
8	1	3	6	4	7	9	5	2

12

2	5	9	7	1	4	3	6	8
8	4	3	5	6	2	1	9	7
6	1	7	9	3	8	2	5	4
7	2	5	3	9	6	4	8	1
4	8	6	1	2	7	5	3	9
9	3	1	8	4	5	7	2	6
3	9	4	2	8	1	6	7	5
5	6	2	4	7	9	8	1	3
1	7	8	6	5	3	9	4	2

13

1	7	6	4	5	2	3	8	9
9	3	4	8	7	1	6	5	2
5	8	2	9	6	3	4	7	1
6	1	7	3	2	8	9	4	5
4	2	8	5	9	7	1	6	3
3	9	5	1	4	6	7	2	8
7	5	9	2	1	4	8	3	6
2	4	3	6	8	9	5	1	7
8	6	1	7	3	5	2	9	4

14

4	3	7	2	9	5	1	8	6
6	5	9	8	3	1	2	4	7
2	1	8	6	4	7	9	3	5
7	2	1	9	6	8	3	5	4
9	6	3	5	2	4	7	1	8
8	4	5	7	1	3	6	2	9
1	9	4	3	5	6	8	7	2
3	7	6	4	8	2	5	9	1
5	8	2	1	7	9	4	6	3

15

2	1	6	9	5	4	7	3	8
3	5	9	7	8	6	1	4	2
7	4	8	3	1	2	5	6	9
6	2	7	1	4	9	8	5	3
5	8	3	6	2	7	4	9	1
1	9	4	5	3	8	2	7	6
8	3	5	4	9	1	6	2	7
9	7	2	8	6	5	3	1	4
4	6	1	2	7	3	9	8	5

16

6	7	4	8	9	3	5	2	1
1	3	9	7	2	5	4	6	8
5	2	8	1	6	4	7	9	3
3	9	2	4	7	8	1	5	6
7	8	1	2	5	6	9	3	4
4	6	5	3	1	9	8	7	2
2	4	6	5	8	7	3	1	9
9	5	3	6	4	1	2	8	7
8	1	7	9	3	2	6	4	5

17

2	4	9	5	1	6	3	8	7
1	7	5	8	9	3	6	4	2
3	8	6	2	7	4	1	9	5
7	6	2	1	5	9	8	3	4
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18

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19

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46

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47

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48

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49

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52

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56

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57

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58

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59

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61

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62

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112

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116

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117

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122

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129

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150

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